Callum Christer

Games Computing student

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Personal Statement

I am a third-year Games Computing student with a strong passion for game development and software engineering. I am currently seeking a junior programming or internship opportunity where I can apply and expand my C++ development skills in a professional environment.

Through my studies, I have built a solid foundation in C++, object-oriented programming, and 3D graphics, alongside experience with Unreal Engine, and collaborative version control. I enjoy solving complex problems and writing efficient, maintainable code to make creative gameplay experiences.

Technical Skills

Languages: C++, PSSL (PlayStation Shader Language)

Software Experience: Visual Studio 2022, Unreal Engine 5.6, Git

Other: OOP, UE5 blueprints, HTML, React, HTML, CSS, Graphics pipelines, Shaders, Mathematics, Problem solving

Current Education

Bachelor's Degree - Games Computing
Northumbria University – Newcastle, UK
Start Date – September 2023
Projected End Date – June 2026
Key Modules – Games Programming 1,2 & 3, Game Design, Computing Consultancy Project, Computational Thinking, Computing Fundamentals, 3D Graphics Programming.

Projects

Cosmos Conqueror (PC) *2025*
Languages: C++
Description: A 2D side-scroller built using C++ in the Hornet game engine. This project showcases effective memory management using raw pointers, utilisation of an event system as well as clean and readable code. The main problem I faced in this project was architecture related caused by tight coupling; I was fighting my own code near the end. To avoid this, I should have made better use of the event system, OOP principles and interfaces.

3D Graphics Application (PS4) *2025/26*
Tools Used: PS4 Dev Kit, Visual Studio, C++, PSSL
Description: A 3D graphics application made for PS4 consoles using PlayStation Shader Language (PSSL). This project showcases a range of graphical techniques and technologies used within the industry. Features include, a variety of 3D models created from model data (position, colour, normals, UVs), a moving camera controlled by the PS4 controller, model transformations, texture mapping, generating smooth and flat normals, multiple pipelines to apply different shader properties to different objects, ADS lighting from multiple light sources including directional, point and spotlight and a very basic HUD that displays text or numbers on the screen that is unaffected by the lighting in the scene.

NPC AI Comparison *2026*
Tools Used: C++, Unreal Engine, Utility AI, Behaviour Trees
Description: A simulation showcasing the impact of behaviour trees and utility AI on NPC decision-making and adaptability in Unreal Engine 5. NPC logic was written in C++ and with the focus of utilising OOP principles such as inheritance and polymorphism. I applied what I learned from Cosmos Conqueror in this project by implementing interfaces and delegates to avoid tight coupling and unnecessary dependencies.